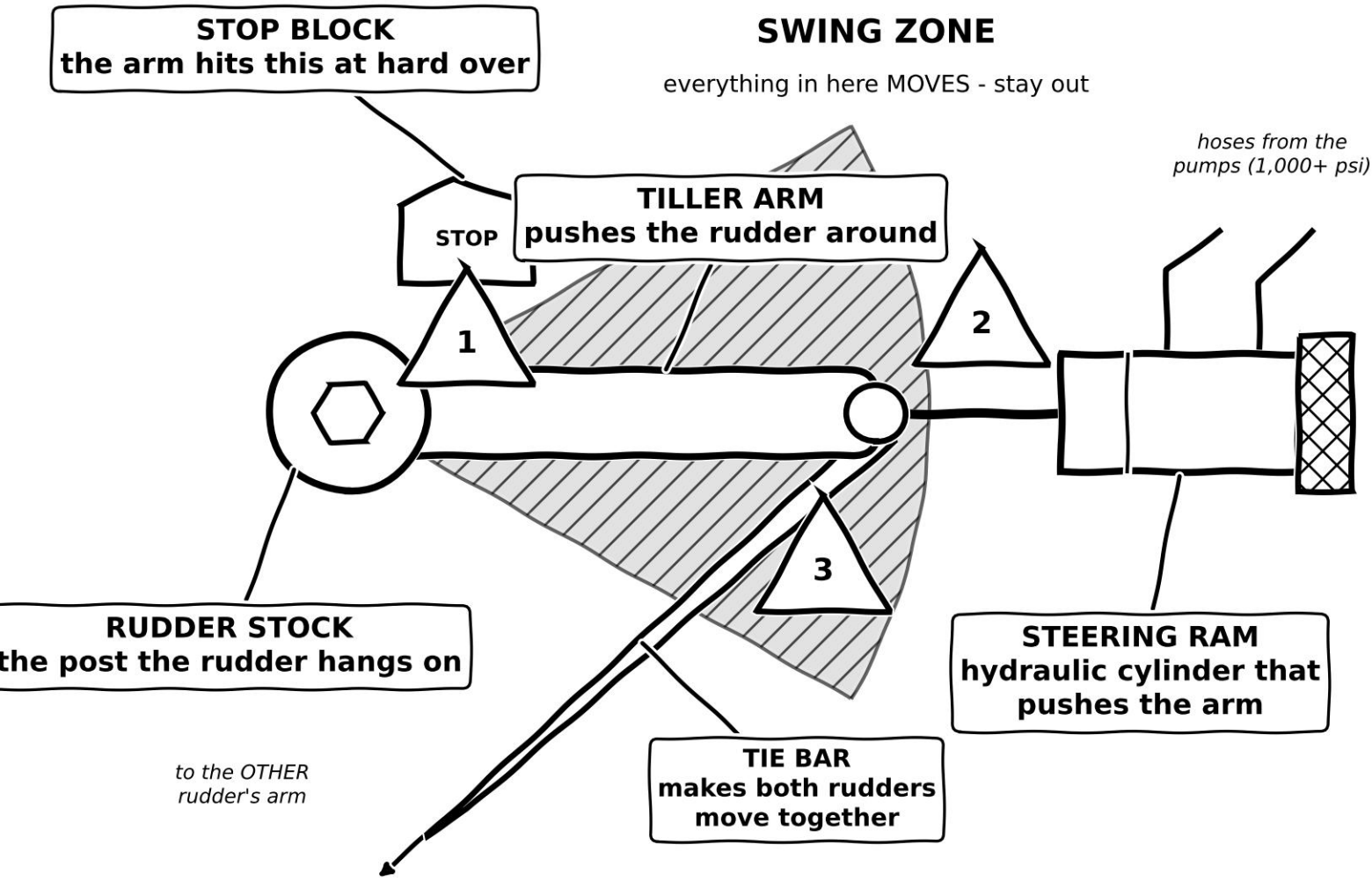
RUDDER ROOM - THE STEERING GEAR

This is ONE rudder, looking straight down at the deck. There's one of these for each steering rudder.



HOW IT WORKS (the short version)

The pilot moves the steering lever. Oil from the pumps pushes the STEERING RAM. The ram pushes the TILLER ARM. The arm turns the RUDDER STOCK, and the rudder in the water turns with it. The TIE BAR drags the other rudder along so both move together.

PINCH POINTS - these will take a hand off

1 Between the ARM and the STOP BLOCK. At hard over the arm slams the block. Nothing survives in between.

2 The RAM ROD and the PIN. The rod slides in and out with no warning, and the pin joint closes like scissors.

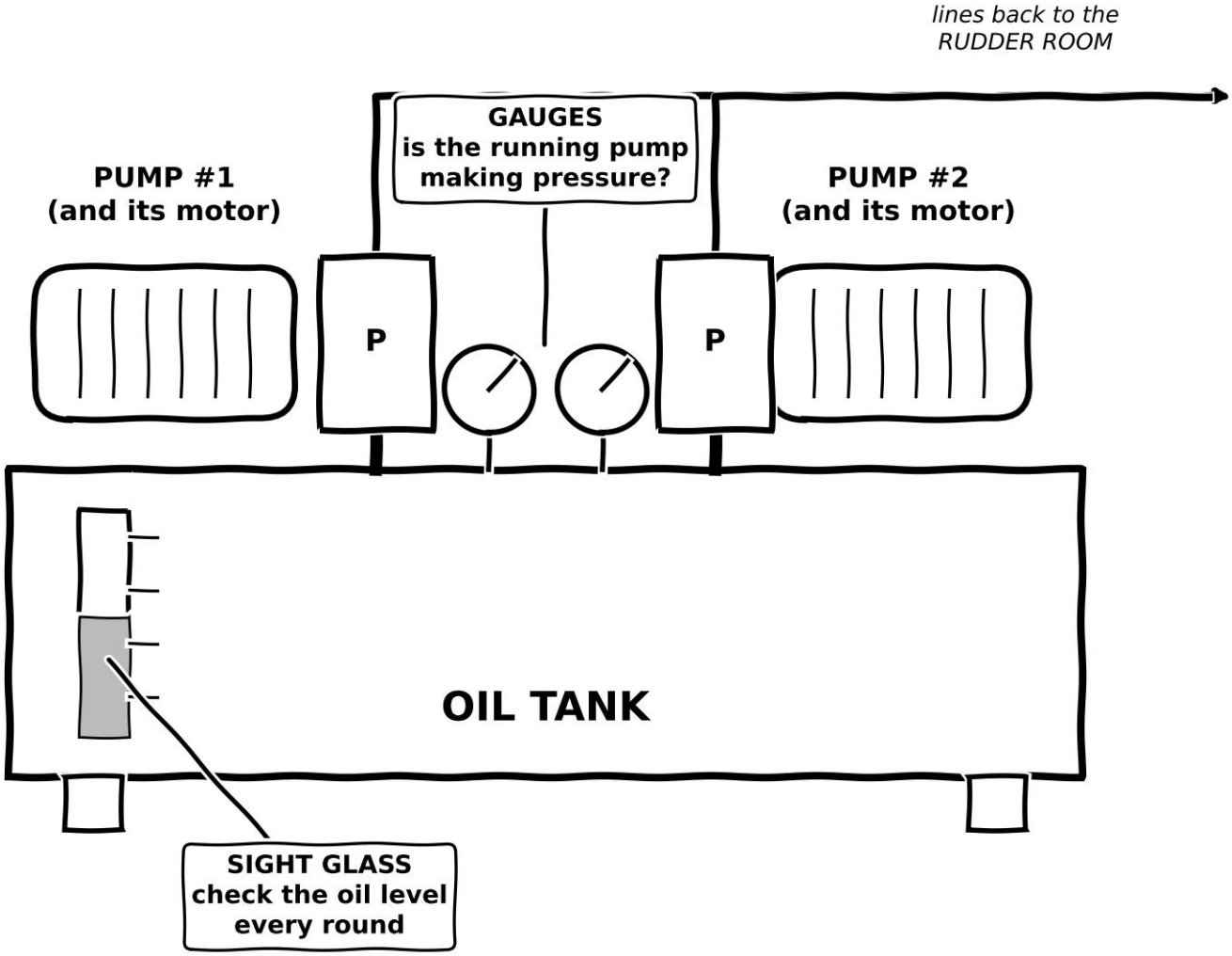
3 The TIE BAR. It swings across the room when the pilot steers. Never step over it, never lean on it.

THE RULES

1. Pumps ON = stay out of the swing zone. The pilot can't see you and doesn't know you're there.
2. Never put a hand on a pin, a rod, or between the arm and the stop.
3. Nobody works in the swing zone until the pilothouse is told and the pumps are OFF and tagged.

ENGINE ROOM - THE STEERING PUMPS

The heart of the system. Two pumps, one tank. If these quit, the boat doesn't steer.



ONE RUNS. ONE IS THE SPARE.

Only one pump runs at a time. The other one sits ready.

WHY WE SWITCH PUMPS EVERY TIME THE BOAT TURNS AROUND

Southbound we run one pump. Turn around northbound, we switch to the other one. Two reasons:

1. Both pumps wear the same, instead of one wearing out while the other sits.
2. The spare gets proved every trip. You never want to find out it's dead with a tow in a bend.

YOUR JOB ON ROUNDS

- Look at the sight glass. If the oil is lower than last time, it went somewhere. Tell the engineer.
- Look under the pumps and the tank for oil on the deck.
- Listen. A pump that sounds different, runs hot, or trips its breaker gets reported right then.
- Don't touch the valves. Know what they are, leave them alone.